

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: Data Structure

Time: 1.5 Hours

Semester / Programme: sem 3

Full Marks: 20

Instructions: Answer Q1 and any 3 from Q2-Q5

SECTION - A (Compulsory)

1(a) What is the time complexity of binary search?

[1]

$$O(\log n)$$

Solution & Key Concept:

- Divide and conquer halves search space
- Recurrence $T(n) = T(n/2) + c$

Answer: $O(\log n)$

Marking Rubric: [1 Mark] for correct bound

SECTION - B

Answer any 3

Q.2. (a) Explain array storage representation.

[2.5]

$$A[i] = B + i \times w$$

Solution for (a):

- Base address plus offset

Result / Conclusion: Computed address

Marking Rubric: [1.5 Marks] calculation, [1 Mark] formula

(b) Derive stack overflow condition.

[2.5]

Solution for (b):

- $TOP == MAX - 1$

Result / Conclusion: Full condition

Marking Rubric: [2.5 Marks] complete description

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